Chalmers University of Technology

Planning of electricity production and transmission

Nonlinear optimization

TMA947 / MMG621

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Abstract

This project investigates electricity production and transmission in a network with 11 nodes and nine generators. The objective is to minimize production cost while meeting demand and accounting for transmission losses. A nonlinear model is formulated using AC power flow equations, active and reactive power balances, generator capacity limits, and bounds on voltage magnitudes and phase angles. The model is implemented in Julia using JuMP and Ipopt. The computed generator outputs, voltages, phase angles, and line flows provide a basis for analysing the resulting dispatch. The cheapest plan found costs 183.24 SEK, with transmission losses of less than 0.1% of the demand. Since the problem is not a convex optimization problem, the solver only guarantees a local optimum; however, using a lower bound from a simpler relaxed problem, we prove that the solution costs at most 0.13% more than the global optimum.

AI disclosure: We used an AI assistant throughout the project to get feedback on our reasoning, help debug and format the Julia code, improve the wording of the report, correct grammar and spelling, and with the LaTeX formatting. We reviewed the feedback from the AI assistent critically and made the final decisions about the model, analysis, and conclusions ourselves. We take full responsibility for the submitted work.

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1 Introduction and Background

Electricity production and transmission must be planned so that consumer demand is met while the power network operates within its technical limits. Generators have different production costs and capacities, and transmitting power through the network causes losses. Consequently, minimizing production cost requires considering both generator outputs and the electrical behaviour of the network. Such methods are also used in industry. For example, Siemens provides "PSS®E", an advanced power system analysis tool supporting generation planning, transmission planning, and optimal power flow calculations [1].

This project studies a simplified network using nonlinear optimization. The model includes active and reactive power flows, generator capacity limits, and bounds on voltage magnitudes and phase angles. The scope is a static operating situation with fixed demand and network parameters, as specified in the project description.

1.1 Problem Description

The network consists of 11 nodes, 15 transmission lines, and nine generators. Demand is distributed across seven nodes, while generators are connected to six nodes. Each generator has a specified production cost and maximum active power capacity. Reactive generation is restricted to $\pm 3\%$ of the generator's maximum active power capacity.

The problem is to determine generator outputs, node voltage magnitudes, and phase angles that minimize total production cost while satisfying active and reactive power balances at every node. Line flows are governed by nonlinear AC power flow equations, and total active generation must cover both demand and transmission losses. Voltage magnitudes are restricted to 0.98–1.02 vu, and node 1 is used as the phase-angle reference.

1.2 Objectives

The objective is to formulate and solve the electricity planning problem in Julia using JuMP and Ipopt. The report aims to determine a feasible dispatch with minimum production cost, report the corresponding generator outputs, voltages, phase angles, and line flows, and quantify total active power losses. Since the model is not a convex optimization problem, the solver only guarantees a local optimum; we therefore also bound how far the solution can be from the global optimum.

2 Mathematical formulation

This section formulates the planning problem as a nonlinear optimization model, known in power engineering as an *optimal power flow* problem. The model determines how much active and reactive power each generator produces, together with the voltage amplitude and phase angle at each node of the network in Figure 1. It does so at the lowest total production cost, while respecting the generators' capacity limits, the allowed voltage ranges, and the requirement that

at every node the power flowing in equals the power flowing out. All data (generator capacities and costs, consumer demands, and line parameters) is taken from Tables 2–4 of the project description.

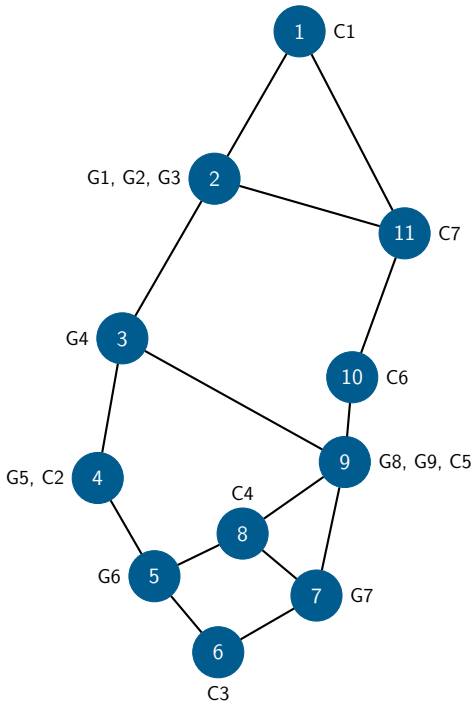


Figure 1: The transmission network: 11 nodes (circles) connected by 15 transmission lines. Labels show the generators G1–G9 and consumers C1–C7 located at each node.

Sets

The set G contains the nine generators, where generator $i \in G$ is labelled Gi in Figure 1, and V contains the eleven nodes. The set E contains the 15 transmission lines connecting the nodes. The set A contains both directions of every line, since each end of a line has its own outgoing flow in its own balance equation, and the active power flows $p_{k\ell}$ and $p_{\ell k}$ generally differ because of transmission losses. For each node $k \in V$, G_k denotes the generators located at that node; for example, $G_2 = \{1, 2, 3\}$, and $G_1 = \emptyset$ since node 1 has no generator. Similarly, N_k denotes the set of nodes directly connected to node k; for example, $N_1 = \{2, 11\}$.

- Generators: $G = \{1, \ldots, 9\}$.
- Nodes: $V = \{1, \ldots, 11\}$.
- Edges, each listed once:

$$\begin{aligned} E = \{&(1,2), (1,11), (2,3), (2,11), (3,4), (3,9), \\ &(4,5), (5,6), (5,8), (6,7), (7,8), (7,9), \\ &(8,9), (9,10), (10,11)\}. \end{aligned}$$

- Directed arcs, containing both orientations of every edge:

$$A = \bigcup_{(k,\ell)\in E} \{(k,\ell),(\ell,k)\}, \qquad |A| = 30.$$

- Generators located at node k: $G_k \subseteq G$, for $k \in V$ (for example $G_2 = \{1,2,3\}$ and $G_1 = \emptyset$).
- Neighbors of node k:

$$N_k = \{\ell \in V : (k,\ell) \in A\}, \qquad k \in V.$$

Parameters

The parameter c_i is the production cost per unit of active power for generator i, and $P_i^{\max}$ is its maximum active power output; both are given data (Table 3 of the project description), not decisions made by the model. The parameter D_k is the active power demand at node k (Table 4), with $D_k = 0$ at nodes without consumers, such as node 2 in Figure 1. The line parameters $b_{k\ell}$ and $g_{k\ell}$ (Table 2) describe how power travels along the line between nodes k and ℓ; in particular, $g_{k\ell}$ determines how much power is lost as heat on the line. Both have the same value in either direction along a line.

- c_i: production cost of generator i [SEK/pu], for $i \in G$.
- $P_i^{\max}$: maximum active power capacity of generator i [pu], for $i \in G$.
- D_k: active power demand at node k [pu], with $D_k = 0$ if the node has no consumer, for $k \in V$.
- $b_{k\ell}, g_{k\ell}$: line parameters for $(k,\ell) \in A$, satisfying $b_{k\ell} = b_{\ell k}$ and $g_{k\ell} = g_{\ell k}$.

Decision variables

The decision variable P_i is the active power produced by generator i, i.e. the power that consumers can actually use. The variable Q_i is the reactive power of generator i: it cannot be used by consumers to extract energy, but it is needed to keep the voltages in the network at the right level. Positive values of Q_i mean production, negative values absorption. The variables v_k and θ_k are the voltage amplitude and phase angle at node k: the amplitude describes the strength of the voltage, and the angle its timing relative to the other nodes. Since the flow formulas (1)–(2) contain the angles only as differences $\theta_k - \theta_\ell$, only these differences, not the individual angles, affect the flows. In total, the model has $9 + 9 + 11 + 11 = 40$ decision variables.

- P_i: active power produced by generator i [pu], for $i \in G$.
- Q_i: reactive power of generator i [pu], for $i \in G$. Positive values indicate production, negative values absorption.
- v_k: voltage amplitude at node k [vu], for $k \in V$.
- θ_k: voltage phase angle at node k [rad], for $k \in V$.

Derived quantities: line flows

For every $(k, \ell) \in A$, the active power $p_{k\ell}$ and reactive power $q_{k\ell}$ flowing out of node k towards node ℓ [pu] are given by

$$p_{k\ell} = v_k^2 g_{k\ell} - v_k v_\ell \left[g_{k\ell} \cos(\theta_k - \theta_\ell) + b_{k\ell} \sin(\theta_k - \theta_\ell) \right], \tag{1}$$

$$q_{k\ell} = -v_k^2 b_{k\ell} + v_k v_\ell \left[b_{k\ell} \cos(\theta_k - \theta_\ell) - g_{k\ell} \sin(\theta_k - \theta_\ell) \right]. \tag{2}$$

These flows are not chosen freely: they are determined by the voltage amplitudes and phase angles at the two ends of the line. This keeps the model at 40 decision variables, instead of adding 60 flow variables (one active and one reactive flow per directed arc). Because lines have losses, the flows in the two directions do not cancel. Their sum is the active power lost on the line as heat, which, as shown in Section 3, satisfies

$$p_{k\ell} + p_{\ell k} = g_{k\ell} \left[v_k^2 + v_\ell^2 - 2 v_k v_\ell \cos(\theta_k - \theta_\ell) \right] \;\geq\; 0.$$

The power leaving one end of a line is therefore always at least as large as the power arriving at the other end.

Optimization problem

$$\min \quad \sum_{i \in G} c_i P_i \tag{3}$$

$$\text{s.t.} \quad 0 \leq P_i \leq P_i^{\max}, \qquad \forall i \in G \tag{4}$$

$$-0.03 P_i^{\max} \leq Q_i \leq 0.03 P_i^{\max}, \qquad \forall i \in G \tag{5}$$

$$0.98 \leq v_k \leq 1.02, \qquad \forall k \in V \tag{6}$$

$$-\pi \leq \theta_k \leq \pi, \qquad \forall k \in V \tag{7}$$

$$\theta_1 = 0, \qquad \text{(reference angle)} \tag{8}$$

$$\sum_{i \in G_k} P_i - D_k = \sum_{\ell \in N_k} p_{k\ell}, \qquad \forall k \in V \tag{9}$$

$$\sum_{i \in G_k} Q_i = \sum_{\ell \in N_k} q_{k\ell}, \qquad \forall k \in V \tag{10}$$

The objective (3) is the total production cost in SEK: for each generator, the price per unit of active power times the amount produced, summed over all generators. Only the active power P_i appears, since the project data assigns a price only to active power; producing reactive power is free, and the voltages have no cost. The variables Q_i, v_k and θ_k still matter, because the constraints below link them to what the generators can deliver. Since they do not appear in the cost, their values need not be unique at the optimum; Section 3 shows that this is the case for how reactive power is split between generators at the same node.

The active power of each generator is bounded by (4): it is non-negative and cannot exceed the generator's capacity. By (5), each generator can produce or absorb reactive power up to 3% of its maximum active power capacity. The voltage amplitudes are kept between 0.98 and 1.02 by

(6), i.e. within ±2% of the nominal value, since electrical equipment is built for a fixed voltage and can malfunction or be damaged if the voltage is too high or too low. Finally, (7) restricts each phase angle to one full turn of the cycle, so that every angle is described in exactly one way.

Since the flow formulas contain the phase angles only as differences, shifting all angles by the same amount leaves every flow unchanged, so without a reference there would be infinitely many equivalent solutions. The constraint (8) removes this ambiguity by fixing $\theta_1 = 0$, much like measuring heights from sea level: moving the zero point changes every number but no difference between them. It therefore affects neither the flows, the production, nor the cost; node 1 is an arbitrary choice.

The active power balance (9) expresses that power cannot appear or disappear at a node: at every node, local production minus local demand must leave the node along its lines. A negative net outflow means that the node imports power from the network. For example, node 1 has no generator and a demand of 0.10 pu, so its balance reads $-0.10 = p_{1,2} + p_{1,11}$: node 1 must import 0.10 pu in total from nodes 2 and 11.

The reactive power balance (10) follows the same principle, but has no demand term: consumers have no use for reactive power, since it cannot be used to extract energy (project description, footnote 4).

In total, the model has 40 decision variables, $23 = 11 + 11 + 1$ equality constraints (active balances, reactive balances and the reference angle), and $80 = 18 + 18 + 22 + 22$ variable bounds (two for each of the variables P_i, Q_i, v_k and θ_k).

3 Results and analysis

Total cost and active power production

The optimal solution has a total production cost of approximately 183.24 SEK, with a total active power production of 0.79070 pu against a total demand of 0.79 pu. The dispatch follows the *merit order*: generators with lower production costs are used first, and seven of the nine generators operate at full capacity (Table 1). Generator G6 is the *marginal* generator: it produces approximately 71% of its capacity to meet the remaining demand and cover the network losses, while the most expensive generator, G7, remains off. Since G6 is only partly used, one additional unit of demand anywhere in the network would cost approximately 350 SEK/pu, the price of G6. The result agrees with a simple merit-order calculation that ignores the network, which gives a cost of 183 SEK; the difference of 0.24 SEK is exactly the cost of G6 covering the transmission losses (0.00070 pu × 350 SEK/pu).

Table 1: Optimal active power production per generator.

Generator	Node	Cost (SEK/pu)	Capacity (pu)	Production (pu)	Utilization (%)
G1	2	175	0.02000	0.02000	100.0
G2	2	100	0.15000	0.15000	100.0
G3	2	150	0.08000	0.08000	100.0
G4	3	150	0.07000	0.07000	100.0
G5	4	300	0.04000	0.04000	100.0
G6	5	350	0.17000	0.12070	71.0
G7	7	400	0.17000	0.00000	0.0
G8	9	300	0.26000	0.26000	100.0
G9	9	200	0.05000	0.05000	100.0
Total			1.01000	0.79070	78.3

Transmission losses and line flows

The active power loss on a line is the sum of the flows in its two directions. Writing out $p_{\ell k}$ from (1), using $g_{\ell k} = g_{k\ell}$ and $b_{\ell k} = b_{k\ell}$, and applying $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$, the sine terms cancel and

$$p_{k\ell} + p_{\ell k} = g_{k\ell}\left[v_k^2 + v_\ell^2 - 2v_k v_\ell \cos(\theta_k - \theta_\ell)\right].$$

Since $\cos(\theta_k - \theta_\ell) \leq 1$ and $v_k, v_\ell > 0$ by the voltage bounds (6), the expression in brackets satisfies

$$v_k^2 + v_\ell^2 - 2v_k v_\ell \cos(\theta_k - \theta_\ell) \;\geq\; v_k^2 + v_\ell^2 - 2v_k v_\ell \;=\; (v_k - v_\ell)^2 \;\geq\; 0.$$

As every line has $g_{k\ell} > 0$ (Table 2 of the project description), $p_{k\ell} + p_{\ell k} \geq 0$ on every line: active power losses are never negative.

The total active power loss is 0.000698 pu, approximately 0.088% of the total demand of 0.79 pu (Table 2). The losses are small because the flows are modest and the voltage amplitudes are kept close to the upper end of their allowed range. The signs in Table 2 show the direction of the flows: for example, $p_{1,2} = -0.09829$ pu means that node 1 receives 0.09829 pu from node 2, while node 2 sends $p_{2,1} = 0.09838$ pu; the difference of 0.00009 pu is lost on the line. The largest losses occur on lines $(8, 9)$ and $(3, 4)$, with 0.000187 pu and 0.000165 pu respectively. Line $(3, 4)$ carries the largest active flow in the network, whereas line $(8, 9)$ carries only the seventh largest; its loss is high because it has the highest resistance of all lines, $r_{k\ell} = g_{k\ell}/(g_{k\ell}^2 + b_{k\ell}^2)$. The loss on a line thus depends both on how much power it carries and on how lossy the line is.

Table 2: Active and reactive power flows in both directions of every line, and the active power loss $p_{k\ell} + p_{\ell k}$ (all in pu). A negative flow means power flowing into node k.

Line (k, ℓ)	$p_{k\ell}$	$p_{\ell k}$	Loss	$q_{k\ell}$	$q_{\ell k}$
(1,2)	−0.09829	0.09838	0.000091	0.00228	−0.00184
(1,11)	−0.00171	0.00171	< 0.000001	−0.00228	0.00228
(2,3)	0.06922	−0.06918	0.000039	−0.00091	0.00118
(2,11)	0.08240	−0.08234	0.000060	0.00010	0.00027
(3,4)	0.11040	−0.11023	0.000165	0.00298	−0.00201
(3,9)	0.02878	−0.02877	0.000007	−0.00235	0.00239
(4,5)	−0.03977	0.03979	0.000020	0.00316	−0.00302
(5,6)	0.05934	−0.05930	0.000038	−0.00154	0.00181
(5,8)	0.02157	−0.02157	0.000007	0.00519	−0.00514
(6,7)	−0.05070	0.05072	0.000014	−0.00181	0.00199
(7,8)	0.01051	−0.01051	0.000002	0.00524	−0.00522
(7,9)	−0.06122	0.06128	0.000056	−0.00626	0.00658
(8,9)	−0.05793	0.05811	0.000187	0.01036	−0.01004
(9,10)	0.00938	−0.00938	0.000001	0.00261	−0.00260
(10,11)	−0.04062	0.04063	0.000011	0.00260	−0.00254
Total			0.000698		

Reactive power

All reactive power outputs lie within their bounds $-0.03P_i^{\max} \le Q_i \le 0.03P_i^{\max}$ (Table 3). Generator G5 operates closest to its limit, at 0.00115 pu or about 96% of its upper bound. Generator G7 produces no active power but still supplies reactive power. The individual reactive power outputs of generators located at the same node are not unique: the reactive balance (10) only involves their sum, and reactive power has no cost, so any split that respects the individual bounds is equally optimal. For example, moving 0.001 pu of reactive power from G2 to G3 gives $Q_2 = -0.00096$ pu and $Q_3 = -0.00164$ pu; both remain within their bounds and the sum at node 2 is unchanged, so the cost and every other variable stay the same. The node totals, −0.00264 pu at node 2 (net absorption) and 0.00154 pu at node 9, are therefore the meaningful results; the individual values in Table 3 are one of infinitely many optimal choices.

Table 3: Reactive power per generator. Positive values mean production, negative values absorption. The last column shows how much of the allowed range $0.03P_i^{\max}$ is used.

Generator	Node	Q_i (pu)	Bound $0.03P_i^{\max}$ (pu)	Share of bound (%)
G1	2	−0.000043	0.00060	7.2
G2	2	−0.001960	0.00450	43.5
G3	2	−0.000641	0.00240	26.7
G4	3	0.001819	0.00210	86.6
G5	4	0.001147	0.00120	95.6
G6	5	0.000633	0.00510	12.4
G7	7	0.000963	0.00510	18.9
G8	9	0.001483	0.00780	19.0
G9	9	0.000057	0.00150	3.8

Voltage amplitudes and phase angles

All voltage amplitudes lie between 1.0172 and 1.0200 vu, near the top of the allowed range (Table 4). This is expected: for the same transmitted power, a higher voltage requires less current, which reduces the losses and hence the extra production by G6 at 350 SEK/pu. At node 2, the largest generation hub, the voltage reaches the upper limit of 1.02 vu exactly, so this bound is binding. Relaxing it cannot increase the cost and would likely lower it slightly. The phase angles range from −0.0092 to 0.0045 rad (at most about 0.5°) relative to the reference $\theta_1 = 0$. Since only the differences between angles affect the flows, the individual values are meaningful only relative to this reference.

Table 4: Voltage amplitude and phase angle at each node. The amplitude at node 2 is at its upper bound.

Node	v_k (vu)	θ_k (rad)
1	1.01915	0
2	**1.02000**	0.00454
3	1.01949	0.00065
4	1.01777	−0.00815
5	1.01801	−0.00463
6	1.01750	−0.00922
7	1.01791	−0.00573
8	1.01716	−0.00678
9	1.01937	−0.00076
10	1.01908	−0.00139
11	1.01926	0.00005

Convexity and optimality of the solution

The objective and the variable bounds are linear, but the balance constraints (9)–(10) are nonlinear equalities, containing products of voltage amplitudes and sines and cosines of angle differences. The problem is therefore not a convex optimization problem, and the guarantee that every local optimum is also global does not apply. The solver Ipopt only guarantees a locally optimal point; it reports such a point as `LOCALLY_SOLVED` regardless of whether the problem is convex, so its status alone says nothing about global optimality.

We can nevertheless bound how far our solution can be from the global optimum. Summing the active balance (9) over all nodes, each line contributes its loss $p_{k\ell} + p_{\ell k} \geq 0$, which gives $\sum_{i \in G} P_i \geq 0.79$ pu: the generators must produce at least the total demand. Every feasible plan satisfies this, so the simpler problem of minimizing the cost subject only to this requirement and the capacity bounds gives a lower bound on the optimal cost. That problem is solved exactly by using the cheapest generators first, which costs 183 SEK. Our solution is feasible (maximum constraint violation $7 \cdot 10^{-11}$) and costs 183.244 SEK, so the global optimal cost C^* satisfies

$$183 \;\leq\; C^* \;\leq\; 183.244 \text{ SEK}.$$

Our solution is thus proven to be at most 0.244 SEK, or 0.13%, more expensive than the global optimum. The lower bound itself cannot be reached: it would require zero losses on every line, which by the loss formula means equal voltages at both ends of every line, and then no power would flow at all, so node 1 could not be supplied.

4 Conclusions

The cheapest way to meet the demand costs approximately 183.24 SEK. The production follows the merit order: the cheaper generators run at full capacity, G6 covers the remaining demand and the losses at 350 SEK/pu, and the most expensive generator, G7, stays off. Transmission losses are small, about 0.088% of the demand, and the voltage amplitudes lie near the upper end of their allowed range. Since the problem is not a convex optimization problem, the solver cannot guarantee a globally optimal solution; however, using a lower bound from a simpler relaxed problem, we proved that our solution costs at most 0.13% more than the global optimum.

The model assumes that consumers have no reactive power demand. This simplifies the problem compared with real networks, where loads such as electric motors do consume reactive power. As a practical next step, the binding voltage limit at node 2 could be examined, since relaxing it would not increase and would likely slightly reduce the production cost.

References

[1] Siemens, *PSS power system simulation and modeling software*. https://www.siemens.com/en-gb/products/pss-software/. Accessed 14 October 2026.

A Julia code

The model was implemented in Julia using the modelling package JuMP and solved with the interior-point solver Ipopt. The file `data.jl` contains the network data, and `main.jl` builds and solves the model of Section 2 and prints the results reported in Section 3.

data.jl

```
# data.jl -- network data from Tables 2-4 of the project description

# ---------- Generators (Table 3) ----------
# Position i in each vector corresponds to generator Gi
gen_node = [2, 2, 2, 3, 4, 5, 7, 9, 9]                          # node of each generator
P_max    = [0.02, 0.15, 0.08, 0.07, 0.04, 0.17, 0.17, 0.26, 0.05] # capacity [pu]
cost     = [175, 100, 150, 150, 300, 350, 400, 300, 200]        # cost [SEK/pu]

# ---------- Demand per node (Table 4) ----------
# Position k corresponds to node k; 0.0 where there is no consumer
D = [0.10, 0.0, 0.0, 0.19, 0.0, 0.11, 0.0, 0.09, 0.21, 0.05, 0.04]   # [pu]

# ---------- Lines (Table 2) ----------
edges  = [(1,2), (1,11), (2,3), (2,11), (3,4), (3,9), (4,5), (5,6),
          (5,8), (6,7), (7,8), (7,9), (8,9), (9,10), (10,11)]
b_list = [-20.1, -22.3, -16.8, -17.2, -11.7, -19.4, -10.8, -12.3,
          -9.2, -13.9, -8.7, -11.3, -7.7, -13.5, -26.7]
g_list = [4.12, 5.67, 2.41, 2.78, 1.98, 3.23, 1.59, 1.71,
          1.26, 1.11, 1.32, 2.01, 4.41, 2.14, 5.06]

# Store b and g for both directions of every line (the arc set A)
b = Dict{Tuple{Int,Int}, Float64}()
g = Dict{Tuple{Int,Int}, Float64}()
for e in 1:length(edges)
    (k, l) = edges[e]
    b[(k, l)] = b_list[e];  b[(l, k)] = b_list[e]
    g[(k, l)] = g_list[e];  g[(l, k)] = g_list[e]
end

# ---------- Sets ----------
nodes = 1:11                    # V
gens  = 1:9                     # G
arcs  = collect(keys(b))        # A: all 30 directed arcs (k, l)

gens_at(k)   = [i for i in gens if gen_node[i] == k]    # G_k
neighbors(k) = [l for (kk, l) in arcs if kk == k]       # N_k
```

main.jl

```julia
# main.jl -- optimal production and transmission planning model

using JuMP, Ipopt
include("data.jl")

model = Model(Ipopt.Optimizer)

# ---------- Decision variables with bounds ----------
@variable(model, 0 <= P[i in gens] <= P_max[i])                    # active power
@variable(model, -0.03*P_max[i] <= Q[i in gens] <= 0.03*P_max[i])  # reactive power
@variable(model, 0.98 <= v[k in nodes] <= 1.02)                    # voltage amplitude
@variable(model, -pi <= theta[k in nodes] <= pi)                   # voltage angle

# ---------- Objective: total production cost ----------
@objective(model, Min, sum(cost[i] * P[i] for i in gens))

# ---------- Active power flow on arc (k,l) ----------
p_flow(k, l) = v[k]^2 * g[(k,l)] -
               v[k]*v[l]*g[(k,l)]*cos(theta[k] - theta[l]) -
               v[k]*v[l]*b[(k,l)]*sin(theta[k] - theta[l])

# ---------- Active power balance at every node ----------
@constraint(model, active_balance[k in nodes],
    sum(P[i] for i in gens_at(k); init = 0) - D[k] ==
    sum(p_flow(k, l) for l in neighbors(k)))

# ---------- Reactive power flow on arc (k,l) ----------
q_flow(k, l) = -v[k]^2 * b[(k,l)] +
               v[k]*v[l]*b[(k,l)]*cos(theta[k] - theta[l]) -
               v[k]*v[l]*g[(k,l)]*sin(theta[k] - theta[l])

# ---------- Reactive power balance at every node ----------
@constraint(model, reactive_balance[k in nodes],
    sum(Q[i] for i in gens_at(k); init = 0) ==
    sum(q_flow(k, l) for l in neighbors(k)))

# ---------- Reference angle ----------
@constraint(model, reference_angle, theta[1] == 0)

# ---------- Solve ----------
optimize!(model)

# ---------- Results ----------
println("Status: ", termination_status(model))
println("Cost: ", objective_value(model))
println("Total production: ", sum(value.(P)))
println("P = ", round.(value.(P), digits = 5))
println("Q = ", round.(value.(Q), digits = 6))
println("v = ", round.(value.(v), digits = 5))
println("theta = ", round.(value.(theta), digits = 5))

println("\nEdge      p_kl      p_lk      p_kl+p_lk   q_kl      q_lk")
for (k, l) in edges
```

```
    pkl = value(p_flow(k, l));  plk = value(p_flow(l, k))
    qkl = value(q_flow(k, l));  qlk = value(q_flow(l, k))
    println("($k,$l)   ", round(pkl, digits=5), "   ", round(plk, digits=5), "   ",
            round(pkl + plk, digits=6), "   ", round(qkl, digits=5), "   ", round(qlk, digits=5)
    )
end
total_loss = sum(value(p_flow(k, l)) + value(p_flow(l, k)) for (k, l) in edges)
println("Total active losses: ", total_loss)
```